GEE 306: RENEWABLE ENERGY SYSTEMS AND TECHNOLOGY

3 Units — 45 Lecture Hours

Comprehensive Lecture Material for Undergraduate Engineering Students

COURSE DESCRIPTION

Renewable Energy Systems and Technology examines conventional and renewable energy resources, methods of extraction and conversion, energy storage and transmission, energy conservation, energy economics, environmental impacts and sustainable energy development.

The course considers fossil fuels, nuclear energy and renewable resources including **solar, biomass, wind, hydropower and geothermal energy**, with particular emphasis on their application to Nigeria and the global energy system.

The central theme of the course is that energy must not only be available and affordable; it must also be **reliable, environmentally responsible, economically viable and socially acceptable**.

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COURSE LEARNING OUTCOMES

At the end of the course, students should be able to:

1. Define energy, power and renewable energy and explain their significance.
2. Classify primary, secondary, final and useful energy.
3. Explain the major stages of an energy conversion system.
4. Describe conventional and renewable energy resources.
5. Explain the extraction and conversion of fossil fuels.
6. Explain nuclear fission and fusion technologies.
7. Analyse solar, wind, biomass, hydro and geothermal energy systems.
8. Perform basic calculations associated with renewable energy systems.
9. Explain energy storage technologies and their applications.
10. Explain electrical energy transmission and distribution.
11. Analyse energy efficiency and conservation measures.
12. Compare different energy technologies from technical, economic, environmental and social perspectives.

13. Explain the concept of sustainable energy and sustainable development.

14. Evaluate energy mixes for Nigeria and other developing economies.

15. Recommend appropriate renewable-energy technologies for different applications.

COURSE ORGANISATION: 45 LECTURE HOURS

Unit	Major Topic	Hours
1	Energy concepts and energy systems	3
2	Global and Nigerian energy systems	3
3	Energy resources and extraction	3
4	Fossil-fuel energy systems	3
5	Nuclear energy systems	3
6	Solar energy	3
7	Biomass energy	3
8	Wind energy	3
9	Hydropower	3
10	Geothermal and other renewable energy	3
11	Energy conversion and power systems	3
12	Energy storage	3
13	Energy transmission, conservation and efficiency	3
14	Energy economics and energy mix	3
15	Sustainable energy and sustainable development	3
Total		**45**

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UNIT 1: INTRODUCTION TO ENERGY SYSTEMS

1.1 Meaning of Energy

Energy is the capacity of a system to perform work or produce a physical change.

The SI unit of energy is the **joule (J)**.

Common units include:

- joule (J)
- kilojoule (kJ)
- megajoule (MJ)
- watt-hour (Wh)
- kilowatt-hour (kWh)
- megawatt-hour (MWh)
- British thermal unit (Btu)
- tonne of oil equivalent (toe)

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The relationship between electrical energy units is:

$$1\,kWh = 3.6\,MJ$$

Power

Power is the rate at which energy is transferred or converted.

$$P = \frac{E}{t}$$

where:

P= power (W)

E= energy (J)

t= time (s)

Therefore:

$$1\,W = 1\,J/s$$

1.2 Energy and Power

A 2 kW electric heater operating for 5 hours consumes:

$$E = Pt$$
$$E = 2 \times 5 = 10\ kWh$$

Therefore:

$$E = 10 \times 3.6 = 36\ MJ$$

Important distinction

Power tells us how fast energy is being used.

Energy tells us how much has been used.

1.3 Types of Energy

Energy can exist in several forms.

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Mechanical energy

Mechanical energy consists of:

- kinetic energy
- potential energy

Kinetic energy:

$$E_k = \frac{1}{2}mv^2$$

Potential energy:

$$E_p = mgh$$

Thermal energy

Associated with temperature and molecular motion.

Chemical energy

Stored in chemical bonds.

Examples:

- petrol

- natural gas
- coal
- biomass
- batteries

Electrical energy

Energy associated with electric charge and electromagnetic fields.

Nuclear energy

Energy stored within atomic nuclei.

Electromagnetic/radiant energy

Energy transported by electromagnetic radiation.

Solar radiation is an important example.

1.4 Primary and Secondary Energy

Primary energy

Energy available directly from natural resources before conversion.

Examples:

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- crude oil
- coal
- natural gas
- sunlight
- wind
- flowing water
- uranium
- biomass

Secondary energy

Energy obtained by converting primary energy.

Examples:

- electricity
- hydrogen
- refined petroleum products

Final energy

Energy delivered to the consumer.

Examples:

- electricity supplied to a factory
- petrol supplied to a vehicle
- LPG supplied to a household

Useful energy

Energy actually providing the desired service.

For example, the useful output of an electric lamp is illumination.

1.5 General Energy Conversion Chain

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Example: Hydroelectric system

Water in reservoir

↓

Potential energy

↓

Water flow

↓

Hydraulic turbine

↓

Mechanical energy

↓

Generator

↓

Electrical energy

↓

Transmission network

↓

Electric motor

↓

Mechanical work

UNIT 2: GLOBAL AND NIGERIAN ENERGY SYSTEMS

2.1 The Global Energy System

The global energy system consists of:

1. Energy resources
2. Extraction systems
3. Conversion technologies
4. Transmission networks
5. Distribution systems
6. Storage systems
7. End-use technologies
8. Energy consumers

The major sectors of energy consumption are:

- electricity
- transportation
- industry

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- residential buildings
- commercial buildings
- agriculture

2.2 Current Global Energy Transition

Renewable energy has become one of the fastest-growing components of the global energy system.

IRENA reports that renewable power capacity reached approximately **4.45 TW at the end of 2024**, representing about 46% of global installed power capacity. Solar accounted for the largest renewable capacity, followed by hydropower and wind. (IRENA)

The expansion continued strongly in 2025. IRENA reports that approximately **692 GW of renewable capacity was added globally in 2025**, taking total renewable capacity to about **5,149 GW**. (IRENA)

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This rapid growth is being driven by:

- declining technology costs
- energy-security concerns
- climate policy
- technological development
- electrification
- energy-access programmes
- private investment
- distributed-generation technologies

2.3 Nigeria's Energy Situation

Nigeria has substantial energy resources including:

Fossil resources

- crude oil
- natural gas

- coal

Renewable resources

- solar
- hydropower
- biomass
- wind
- geothermal potential

Nigeria's energy challenge is unusual because the country possesses substantial natural resources while still facing major challenges relating to **energy access, reliability, affordability and infrastructure**.

2.4 Nigeria Energy Transition Plan

Nigeria has adopted an Energy Transition Plan aimed at achieving **net-zero emissions by 2060** while simultaneously addressing energy poverty and economic development. The plan covers five major sectors:

- power
- cooking
- transportation
- industry
- oil and gas

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The Nigerian Energy Transition Plan explicitly links energy transition with improved energy access, poverty reduction, modern energy services and employment. (Nigeria Energy Transition Plan)

An important feature of Nigeria's transition strategy is that natural gas is considered a transition fuel while renewable technologies are expanded. (Nigeria Energy Transition Plan)

2.5 Nigeria's Renewable Energy Opportunities

Different parts of Nigeria have different renewable-energy characteristics.

Resource **Major opportunity**

Resource	Major opportunity
Solar	Nationwide, particularly northern Nigeria
Hydropower	Rivers and high-head locations
Biomass	Agricultural and forestry residues
Wind	Selected northern/coastal locations
Geothermal	Potential requiring further exploration
Waste-to-energy	Major cities and industrial centres

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UNIT 3: ENERGY RESOURCES AND EXTRACTION

3.1 Classification of Energy Resources

Energy resources can be classified as:

Renewable

Resources naturally replenished over relatively short periods.

Examples:

- solar
- wind
- hydro
- biomass
- geothermal

Non-renewable

Resources that exist in finite quantities and are depleted through use.

Examples:

- coal
- oil
- natural gas
- uranium

3.2 Renewable versus Non-Renewable Energy

Characteristic	Renewable	Non-renewable
Resource availability	Replenished	Finite
Examples	Solar, wind, hydro	Coal, oil, gas
Emissions	Generally low during operation	Generally high for fossil fuels
Fuel cost	Usually very low/zero	Significant
Variability	Some are variable	Generally dispatchable
Storage need	Often significant	Usually lower
Environmental impact	Generally lower	Can be substantial

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3.3 Energy Resource Extraction

Extraction means obtaining an energy resource from nature.

Oil extraction

Exploration

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Drilling

↓

Production

↓

Separation

↓

Transportation

↓

Refining

↓

Petroleum products

Coal

Exploration

↓

Mining

↓

Crushing

↓

Processing

↓

Transportation

↓

Combustion

↓

Heat

↓

Electricity

Natural gas

Gas reservoir

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Drilling

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Gas processing

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Pipeline / LNG

↓

Gas turbine

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Generator

↓

Electricity

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UNIT 4: FOSSIL-FUEL ENERGY SYSTEMS

4.1 Coal

Coal contains stored chemical energy formed from ancient organic matter.

Major types include:

- lignite
- sub-bituminous coal
- bituminous coal
- anthracite

4.2 Coal-Fired Power Plant

COAL

↓

Coal crusher

↓

Boiler

↓

Steam energy

↓

Turbine

↓

Mechanical power

↓

Generator

↓

ELECTRICITY

The basic conversion is:

$$Chemical \rightarrow Thermal \rightarrow Mechanical \rightarrow Electrical$$

4.3 Natural Gas

Natural gas consists mainly of methane:

$$CH_4$$

Its combustion can be represented approximately by:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{heat}$$

4.4 Gas Turbine Power Plant

Air ———→ Compressor

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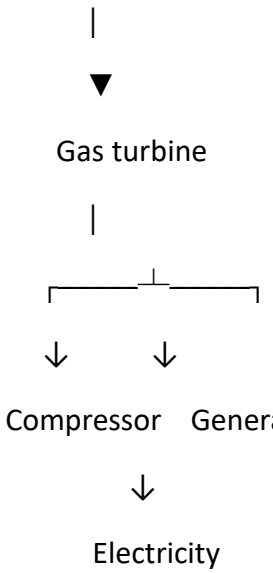
Fuel ———→ Combustor

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Hot high-pressure gas

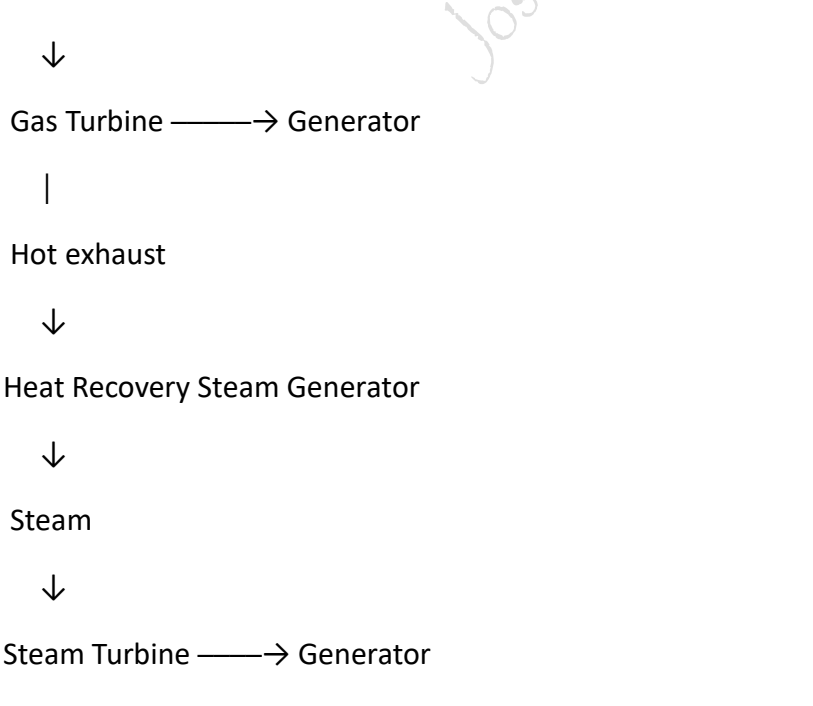
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4.5 Combined-Cycle Gas Turbine

A combined-cycle plant combines:

1. Gas turbine
2. Heat recovery steam generator
3. Steam turbine



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This arrangement improves overall efficiency.

4.6 Advantages of Fossil Fuels

- high energy density
- established infrastructure
- dispatchable
- relatively easy transportation
- suitable for large-scale power generation

Disadvantages

- greenhouse-gas emissions
- air pollution
- finite resources
- environmental degradation
- oil-spill risks
- mining impacts
- price volatility

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SOLVED EXAMPLE 1: POWER-PLANT EFFICIENCY

A thermal power station receives 1,000 MJ of chemical energy from fuel and produces 350 MJ of electrical energy.

Calculate the efficiency.

Solution

$$\eta = \frac{E_{out}}{E_{in}} \times 100$$
$$\eta = \frac{350}{1000} \times 100$$
$$\boxed{\eta = 35\%}$$

The remaining 65% is lost mainly through heat rejection, exhaust gases, mechanical losses and auxiliary consumption.

UNIT 5: NUCLEAR ENERGY

5.1 Nuclear Energy

Nuclear energy is released from changes occurring within an atomic nucleus.

Two major nuclear processes are:

1. Nuclear fission
2. Nuclear fusion

5.2 Nuclear Fission

Fission involves splitting a heavy nucleus into smaller nuclei.

A simplified reaction is:

$$^{235}U + n \rightarrow \text{fission products} + 2 \text{ or } 3n + \text{energy}$$

The emitted neutrons can cause additional fission reactions.

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5.3 Nuclear Chain Reaction

Neutron

↓

Uranium-235

↓

FISSION

↙ ↓ ↘

neutron neutron neutron

↓ ↓ ↓

fission fission fission

↓ ↓ ↓

└—— Chain reaction ——┘

A controlled chain reaction is maintained inside a nuclear reactor.

5.4 Nuclear Power Plant

Nuclear Reactor

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Heat

↓

Steam Generator

↓

Steam

↓

Steam Turbine

↓

Generator

↓

Electricity

Thus:

$$Nuclear \rightarrow Thermal \rightarrow Mechanical \rightarrow Electrical$$

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5.5 Nuclear Fusion

Fusion involves combining light atomic nuclei to form a heavier nucleus.

One commonly studied reaction is:

$$^{2}H + \ ^{3}H \rightarrow \ ^{4}He + n + \text{energy}$$

Fusion has potentially enormous energy density but requires extremely high temperatures and sophisticated confinement systems.

5.6 Advantages and Disadvantages

Advantages

- high energy density
- low direct carbon emissions during operation
- reliable baseload electricity

- small fuel requirement

Disadvantages

- radioactive waste
- high capital cost
- complex safety requirements
- long construction periods
- accident consequences
- nuclear proliferation concerns

UNIT 6: SOLAR ENERGY

6.1 Solar Energy

The Sun is the principal source of energy driving many natural processes on Earth.

Solar technologies can be classified into:

Solar thermal

Converts solar radiation into heat.

Solar photovoltaic

Converts sunlight directly into electricity.

6.2 Solar Photovoltaic Effect

A photovoltaic cell converts solar radiation into electrical energy through semiconductor processes.

Typical materials include:

- silicon
- thin-film materials
- emerging perovskite materials

6.3 PV System

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Sunlight

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PV Modules

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DC Electricity

↓

Charge Controller

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Battery ————┐

↓ |

Inverter ←————┘

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AC Electricity

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Loads

A grid-connected system may omit the battery.

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6.4 Solar Radiation

Solar irradiance is commonly measured in:

$$W/m^2$$

Solar energy incident on a surface is:

$$E = GAt$$

where:

G= solar irradiance

A= area

t= time

SOLVED EXAMPLE 2: SOLAR PV OUTPUT

A PV module has an area of $10m^2$, receives an average irradiance of $800W/m^2$, and has an efficiency of 20%.

Calculate its electrical power output.

Solution

Incident solar power:

$$P_{solar} = GA$$
$$P_{solar} = 800 \times 10$$
$$P_{solar} = 8000W$$

Electrical output:

$$P_{out} = \eta P_{solar}$$
$$P_{out} = 0.20(8000)$$
$$\boxed{P_{out} = 1600W}$$

Therefore, the array produces approximately:

$$\boxed{1.6kW}$$

under the stated conditions.

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6.5 Solar Thermal Systems

Solar thermal technologies include:

- solar water heaters
- solar dryers
- solar cookers
- solar air heaters
- concentrated solar power

Basic solar thermal system

Sun

↓

Solar Collector

↓

Heated Fluid

↓

Heat Exchanger

↓

Hot Water / Heat

6.6 Advantages of Solar Energy

- abundant
- no fuel cost
- modular
- suitable for rural areas
- scalable from watts to utility scale
- low operating emissions

Limitations

- intermittent
- night-time unavailability
- weather dependence
- land requirements
- storage requirements
- inverter and battery costs

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UNIT 7: BIOMASS ENERGY

7.1 Meaning of Biomass

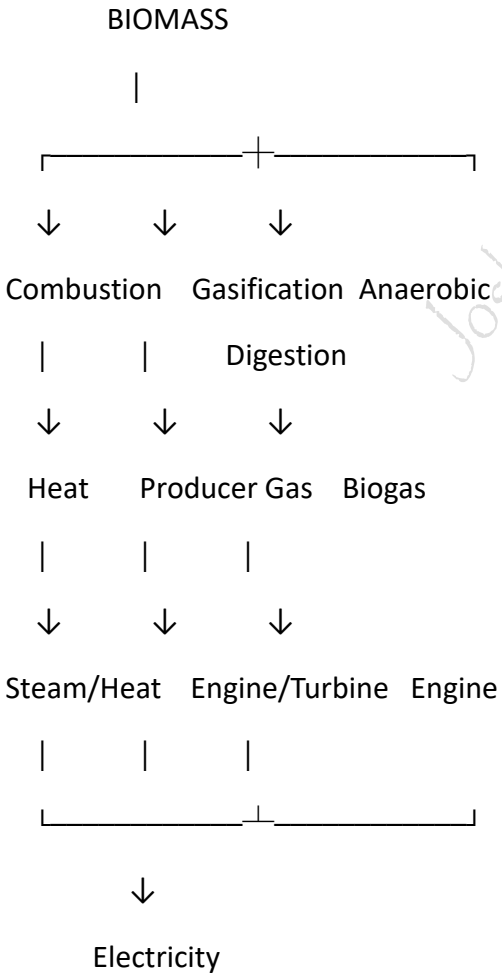
Biomass is organic material that can be used as an energy resource.

Examples include:

- wood
- agricultural residues
- animal waste
- municipal organic waste
- energy crops
- sawdust
- rice husks
- palm residues

Biomass is particularly important in agricultural economies.

7.2 Biomass Energy Conversion Routes



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7.3 Biomass Combustion

Biomass can be directly burned to produce heat.

For electricity:

$$Biomass \rightarrow Heat \rightarrow Steam \rightarrow Turbine \rightarrow Generator$$

7.4 Biogas

Organic matter can undergo anaerobic digestion to produce biogas.

The major component is methane.

Animal manure

Food waste

Sewage

↓

Anaerobic digester

↓

Biogas

↓

Gas cleaning

↓

Engine

↓

Generator

↓

Electricity

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7.5 Biomass Advantages

- uses waste materials
- potentially renewable
- can provide dispatchable power

- supports rural development
- creates employment
- reduces uncontrolled waste disposal

Problems

- feedstock availability
- transport cost
- land competition
- air emissions
- moisture content
- sustainability concerns if forests are overexploited

SOLVED EXAMPLE 3: BIOMASS ENERGY

A biomass plant receives 2,000 kg of dry biomass per hour with a heating value of 15 MJ/kg. The overall electrical efficiency is 25%.

Determine the electrical power output.

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Solution

Thermal energy input per hour:

$$E_{in} = 2000 \times 15$$
$$E_{in} = 30{,}000 MJ/h$$

Convert to MW:

$$1MW = 3600MJ/h$$

Therefore:

$$P_{in} = \frac{30000}{3600} = 8.333MW$$

Electrical power:

$$P_{out} = 0.25(8.333)$$
$$\boxed{P_{out} = 2.08MW}$$

UNIT 8: WIND ENERGY

8.1 Wind Energy Principle

Wind contains kinetic energy because moving air possesses velocity.

The available power in wind is:

$$P_w = \frac{1}{2}\rho A v^3$$

where:

- ρ= air density (kg/m^3)
- A= swept area (m^2)
- v= wind velocity (m/s)

For a horizontal-axis wind turbine:

$$A = \pi R^2$$

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8.2 Wind Turbine

Wind

→→→→

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\|/

[Blades]

|

Rotor

|

Low-speed shaft

|

Gearbox*

|

Generator

|

Converter

|

Electricity

∗Some modern turbines use direct-drive generators.

8.3 Power Coefficient

A turbine cannot extract all the kinetic energy from wind.

$$P_t = C_p\left(\frac{1}{2}\rho A v^3\right)$$

where $C_p$is the power coefficient.

The theoretical Betz limit is approximately:

$$C_{p,max} = 0.593$$

Practical turbines operate below this value.

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SOLVED EXAMPLE 4: WIND TURBINE POWER

A wind turbine has rotor diameter of 20 m. Wind speed is 8 m/s, air density is $1.225 kg/m^3$, and $C_p = 0.40$.

Determine the mechanical power extracted.

Solution

Swept area:

$$A = \frac{\pi D^2}{4}$$
$$A = \frac{\pi(20)^2}{4}$$
$$A = 314.16 m^2$$

Available wind power:

$$P_w = \frac{1}{2}(1.225)(314.16)(8)^3$$
$$P_w \approx 98{,}425W$$

Extracted power:

$$P_t = 0.4(98{,}425)$$
$$\boxed{P_t \approx 39.37kW}$$

8.4 Tip-Speed Ratio

The tip-speed ratio is:

$$\lambda = \frac{\omega R}{v}$$

where:

- ω= rotor angular velocity
- R= rotor radius
- v= wind velocity

This is an important parameter in turbine control.

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UNIT 9: HYDROPOWER

9.1 Hydropower Principle

Hydropower converts the potential and/or kinetic energy of water into mechanical and electrical energy.

The ideal hydraulic power is:

$$P = \rho g Q H$$

Practical output:

$$P_e = \eta \rho g Q H$$

where:

- ρ= water density

- g= gravitational acceleration
- Q= flow rate
- H= hydraulic head
- η= overall efficiency

9.2 Hydroelectric Power Plant

Reservoir

|

| H

↓

Penstock

|

↓

Turbine

|

Mechanical

power

↓

Generator

↓

Transformer

↓

Transmission

9.3 Types of Hydropower

Impoundment

Uses a dam and reservoir.

Run-of-river

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Uses natural river flow with little storage.

Pumped-storage

Stores energy by pumping water to an upper reservoir.

SOLVED EXAMPLE 5: HYDROPOWER

A hydroelectric plant has:

- $Q = 10m^3/s$
- $H = 50m$
- $\rho = 1000kg/m^3$
- $\eta = 0.85$

Calculate output power.

Solution

$$P = \eta\rho gQH$$
$$P = 0.85(1000)(9.81)(10)(50)$$
$$P = 4{,}169{,}250W$$

Therefore:

$$\boxed{P \approx 4.17MW}$$

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9.4 Small and Micro Hydropower

Classification can vary by country, but generally:

- pico hydro: very small
- micro hydro: small community systems
- mini hydro: larger small-scale systems
- small hydro: larger installations

Small hydro is particularly attractive for rural electrification where suitable rivers exist.

UNIT 10: GEOTHERMAL AND OTHER RENEWABLE ENERGY

10.1 Geothermal Energy

Geothermal energy is heat obtained from the Earth's interior.

Potential applications include:

- electricity generation
- district heating
- greenhouse heating
- industrial processes
- bathing

10.2 Geothermal Power Plant

Earth's heat

↓

Hot underground reservoir

↓

Production well

↓

Steam / hot water

↓

Turbine

↓

Generator

↓

Electricity

↓

Injection well

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↓

Geothermal reservoir

Types include:

- dry steam
- flash steam
- binary-cycle systems

10.3 Ocean Energy

Ocean energy includes:

- tidal energy
- wave energy
- ocean thermal energy conversion
- tidal-stream energy

These technologies remain less widely deployed than solar, wind and hydropower.

10.4 Renewable Energy Comparison

Technology	Main energy form	Major advantage	Major challenge
Solar PV	Radiation	Modular	Intermittency
Solar thermal	Radiation	Useful heat	Weather
Wind	Kinetic	Low operating emissions	Variable wind
Hydro	Potential/kinetic	Dispatchable	Site/environment
Biomass	Chemical	Dispatchable	Feedstock
Geothermal	Thermal	Continuous	Resource location
Tidal	Kinetic/potential	Predictable	High cost/site constraints

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UNIT 11: ENERGY CONVERSION SYSTEMS

11.1 Energy Conversion

Energy conversion means changing energy from one form into another.

Examples:

$$Chemical \rightarrow Thermal$$
$$Thermal \rightarrow Mechanical$$
$$Mechanical \rightarrow Electrical$$
$$Solar \rightarrow Electrical$$
$$Hydraulic \rightarrow Mechanical$$

11.2 Energy Conversion Efficiency

$$\eta = \frac{Useful\ Output}{Energy\ Input}$$

For a system containing several conversion stages:

$$\eta_{overall} = \eta_1 \eta_2 \eta_3 \cdots \eta_n$$

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SOLVED EXAMPLE 6: CASCADED EFFICIENCY

A renewable-energy system has:

- turbine efficiency = 90%
- generator efficiency = 95%
- converter efficiency = 96%

Calculate overall efficiency.

$$\eta_o = 0.90 \times 0.95 \times 0.96$$
$$\eta_o = 0.8208$$

Therefore:

$$\boxed{\eta_o = 82.08\%}$$

This demonstrates an important engineering principle: **small losses at several stages can produce a significant overall loss.**

11.3 Energy Conversion Comparison

System	Input	Conversion	Output
PV	Solar radiation	Semiconductor conversion	Electricity
Wind	Air kinetic energy	Rotor/generator	Electricity
Hydro	Water energy	Turbine/generator	Electricity
Coal	Chemical	Boiler/turbine/generator	Electricity
Gas	Chemical	Combustion/turbine	Electricity
Biomass	Chemical	Thermal/biochemical	Heat/electricity
Geothermal	Earth heat	Turbine/generator	Electricity

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UNIT 12: ENERGY STORAGE

12.1 Why Energy Storage is Necessary

Renewable energy sources such as solar and wind are often variable.

Energy storage can:

- balance supply and demand
- provide backup
- improve grid stability
- reduce renewable curtailment
- provide frequency regulation
- support off-grid systems
- improve energy reliability

12.2 Major Energy Storage Technologies

Electrochemical

- lead-acid batteries
- lithium-ion batteries
- sodium-ion batteries
- flow batteries

Mechanical

- pumped hydro
- compressed-air energy storage
- flywheels

Thermal

- molten salt
- hot water
- phase-change materials

Chemical

- hydrogen
- synthetic fuels

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12.3 Battery Energy

The approximate stored energy of a battery is:

$$E = VIt$$

For a battery:

$$E_{Wh} = V \times Ah$$

SOLVED EXAMPLE 7: BATTERY STORAGE

A 24 V, 200 Ah battery is used in an energy-storage system.

Calculate its nominal energy capacity.

$$E = V \times Ah$$
$$E = 24 \times 200$$
$$\boxed{E = 4800Wh}$$

or:

$$\boxed{E = 4.8kWh}$$

If only 80% is usable:

$$E_{usable} = 0.8(4.8)$$
$$\boxed{E_{usable} = 3.84kWh}$$

12.4 Pumped-Hydro Storage

Upper Reservoir

|

| Water

↓

Turbine

|

Generator

|

Electricity

|

Pumping system

↑

|

Lower Reservoir

During periods of surplus electricity, water is pumped upward.

During high demand, water flows downward through the turbine.

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12.5 Hydrogen Energy Storage

A simplified pathway is:

Renewable Electricity

↓

Electrolyser

↓

Hydrogen

↓

Compression / Storage

↓

Fuel Cell / Turbine

↓

Electricity

Hydrogen is therefore an **energy carrier**, rather than a primary energy resource.

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UNIT 13: ENERGY TRANSMISSION, CONSERVATION AND EFFICIENCY

13.1 Electrical Transmission

Electricity is transmitted at high voltage because increasing voltage reduces current for a given power.

For three-phase systems:

$$P = \sqrt{3}VI\cos\phi$$

Therefore:

$$I = \frac{P}{\sqrt{3}V\cos\phi}$$

Transmission-line losses are approximately:

$$P_{loss} = I^2R$$

SOLVED EXAMPLE 8: TRANSMISSION LOSS

A 10 MW load is supplied at 100 kV with power factor 0.9.

Calculate the line current.

$$I = \frac{P}{\sqrt{3}Vpf}$$
$$I = \frac{10 \times 10^6}{\sqrt{3}(100 \times 10^3)(0.9)}$$
$$\boxed{I \approx 64.15A}$$

If the line resistance is 20 Ω:

$$P_{loss} = I^2R$$
$$P_{loss} = (64.15)^2(20)$$
$$\boxed{P_{loss} \approx 82.3kW}$$

This illustrates why high-voltage transmission is important.

13.2 Energy Conservation

Energy conservation means reducing unnecessary energy use while maintaining the required service.

Examples:

- efficient lighting
- efficient motors
- insulation
- efficient air-conditioning
- power-factor improvement
- efficient transformers
- energy-management systems

- efficient industrial processes

13.3 Energy Efficiency

Energy efficiency is:

$$\eta = \frac{Useful\ Energy\ Output}{Energy\ Input}$$

An energy-efficient device provides the same or better service with less energy.

Example

Replacing an incandescent lamp with an efficient LED reduces electricity consumption while providing comparable illumination.

13.4 Energy Conservation versus Energy Efficiency

Energy conservation focuses on avoiding unnecessary energy use.

Energy efficiency focuses on obtaining the same service with less energy.

Example:

- Switching off an unnecessary lamp = conservation.
- Replacing a 100 W lamp with a 15 W LED = efficiency.

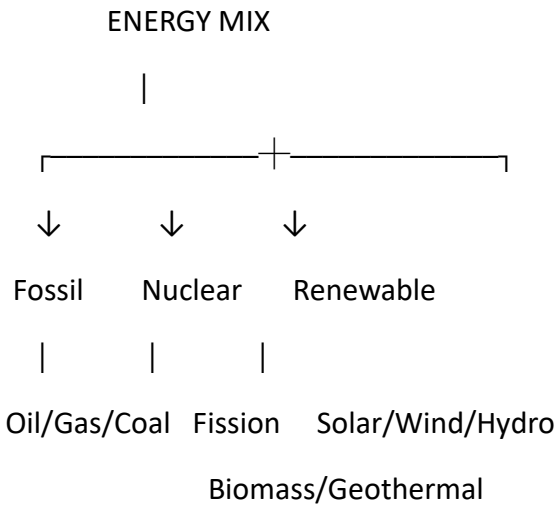
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UNIT 14: ENERGY ECONOMICS AND ENERGY MIX

14.1 Energy Mix

An energy mix is the combination of different energy sources used by a country, region, industry or facility.

For example:



14.2 Why Energy Mix Matters

A country should not normally depend excessively on one energy source.

A diversified energy mix can improve:

- energy security
- reliability
- affordability
- resilience
- economic stability

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14.3 Technical Factors in Energy-Mix Analysis

Engineers consider:

- installed capacity
- energy resource availability
- capacity factor
- reliability
- dispatchability
- transmission capacity
- storage requirements
- system stability

- geographical distribution

14.4 Economic Factors

Important economic indicators include:

- capital cost
- operating cost
- fuel cost
- maintenance cost
- financing cost
- levelized cost of electricity
- payback period
- net present value
- internal rate of return

14.5 Levelized Cost of Electricity

A simplified expression is:

$$LCOE = \frac{\text{Lifetime discounted costs}}{\text{Lifetime discounted electricity generated}}$$

It allows different generating technologies to be compared on an approximately common basis.

SOLVED EXAMPLE 9: SIMPLE LCOE

A solar project costs ₦100 million and is expected to produce 2,000 MWh annually for 20 years. Ignore discounting, replacement and operating costs for this simplified calculation.

Total lifetime generation:

$$E = 2000 \times 20$$
$$E = 40{,}000MWh$$

Therefore:

$$LCOE = \frac{100{,}000{,}000}{40{,}000}$$
$$\boxed{LCOE = ₦2{,}500/MWh}$$

or:

$$\boxed{₦2.50/kWh}$$

This is only a simplified illustration; real project LCOE must account for discount rates, financing, operation and maintenance, degradation, replacements and other costs.

14.6 Capacity Factor

Capacity factor is:

$$CF = \frac{\text{Actual energy generated}}{\text{Maximum possible energy}}$$

For a year:

$$CF = \frac{E_{actual}}{P_{rated} \times 8760}$$

SOLVED EXAMPLE 10: CAPACITY FACTOR

A 5 MW solar plant generates 8,000 MWh per year.

Maximum possible generation:

$$E_{max} = 5 \times 8760$$
$$E_{max} = 43{,}800MWh$$

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Therefore:

$$CF = \frac{8000}{43800}$$
$$\boxed{CF = 18.26\%}$$

UNIT 15: SUSTAINABLE ENERGY AND SUSTAINABLE DEVELOPMENT

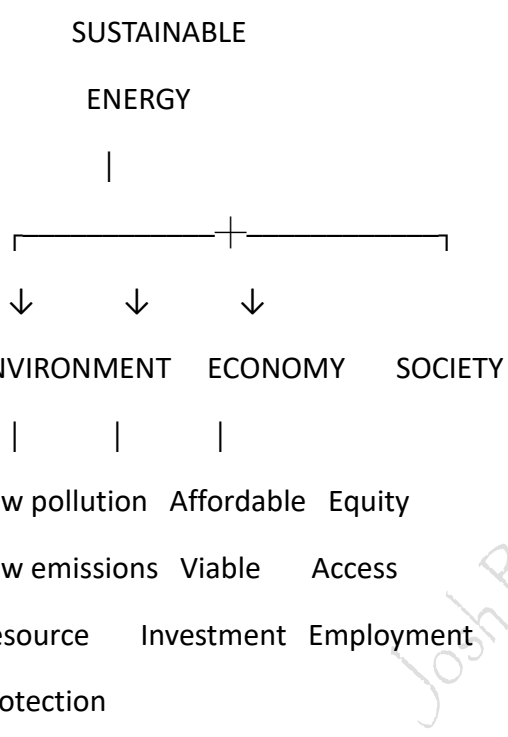
15.1 Meaning of Sustainability

Sustainability means meeting present needs without undermining the ability of future generations to meet their own needs.

Sustainable energy therefore requires an energy system that is:

- environmentally responsible
- economically viable
- socially inclusive
- technically reliable

15.2 Three Pillars of Sustainability



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15.3 Environmental Dimension

Sustainable energy should minimise:

- greenhouse-gas emissions
- air pollution
- water pollution
- land degradation
- biodiversity loss
- hazardous waste

Renewable energy can substantially reduce dependence on fossil fuels, although renewable projects themselves can have environmental impacts.

For example:

- hydropower can alter river ecosystems
- solar farms require land and materials
- wind turbines can affect landscapes and wildlife
- biomass can cause air emissions if poorly managed

Therefore, **renewable does not automatically mean impact-free**.

15.4 Economic Dimension

A sustainable energy system must be financially viable.

It should consider:

- affordability
- investment
- local manufacturing
- job creation
- operation and maintenance
- fuel-price risks
- energy security

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15.5 Social Dimension

Energy sustainability requires equitable access to modern energy services.

Important issues include:

- energy poverty
- rural electrification
- affordability
- employment
- health
- gender impacts
- community participation

15.6 Renewable Energy and Sustainable Development

Renewable energy contributes to several Sustainable Development Goals (SDGs), especially:

SDG 7

Affordable and Clean Energy

SDG 9

Industry, Innovation and Infrastructure

SDG 11

Sustainable Cities and Communities

SDG 12

Responsible Consumption and Production

SDG 13

Climate Action

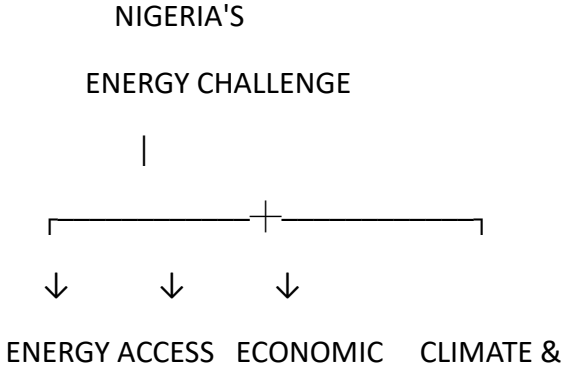
Renewable energy can also indirectly support:

- education
- healthcare
- employment
- poverty reduction
- industrial development

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15.7 Sustainable Energy in Nigeria

Nigeria's energy transition must balance three major objectives:



DEVELOPMENT ENVIRONMENT

The Nigerian Energy Transition Plan seeks to combine climate action with economic development and improved access to modern energy services, with a 2060 net-zero pathway. (Nigeria Energy Transition Plan)

PART II: COMPARATIVE ANALYSIS OF ENERGY TECHNOLOGIES

16.1 Comprehensive Comparison

Parameter	Coal	Gas	Nuclear	Solar	Wind	Hydro	Biomass
Renewable	No	No	No	Yes	Yes	Yes	Yes*
Fuel required	Yes	Yes	Yes	No	No	No	Yes
Direct operational CO_2	High	High	Very low	Very low	Very low	Very low	Variable
Dispatchable	Yes	Yes	Yes	No	No	Yes	
Storage need	Low	Low	Low	High	High	Reservoir	Low
Fuel-price exposure	High	High	Low	Very low	Very low	Very low	
Site dependence	Moderate	Moderate	Moderate	Moderate	High	High	High
Waste issues	High	Moderate	Radioactive	Material waste	Material waste	Sediment/ecology	Ash/emissions

*Biomass is renewable only when its feedstock is sustainably managed.

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PART III: ENERGY SYSTEMS AND END-USE TECHNOLOGIES

17.1 Energy End-Use Sectors

Residential

- lighting
- refrigeration
- cooking
- cooling
- water heating
- electronics

Industrial

- motors
- heating
- pumping
- compressed air
- process systems

Transportation

- petrol/diesel vehicles
- electric vehicles
- hydrogen vehicles
- rail systems

Agriculture

- irrigation
- processing
- drying
- refrigeration

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17.2 Electrification of Energy Services

One important feature of the modern energy transition is electrification.

Examples:

$$Petrol\ vehicle \rightarrow Electric\ vehicle$$
$$Gas\ heating \rightarrow Heat\ pump$$
$$Diesel\ pump \rightarrow Solar\ electric\ pump$$

The electricity can increasingly be generated using renewable resources.

PART IV: INTEGRATED RENEWABLE ENERGY SYSTEMS

18.1 Hybrid Energy Systems

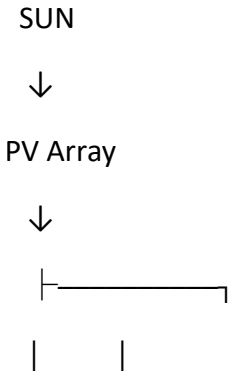
A hybrid energy system combines two or more energy resources.

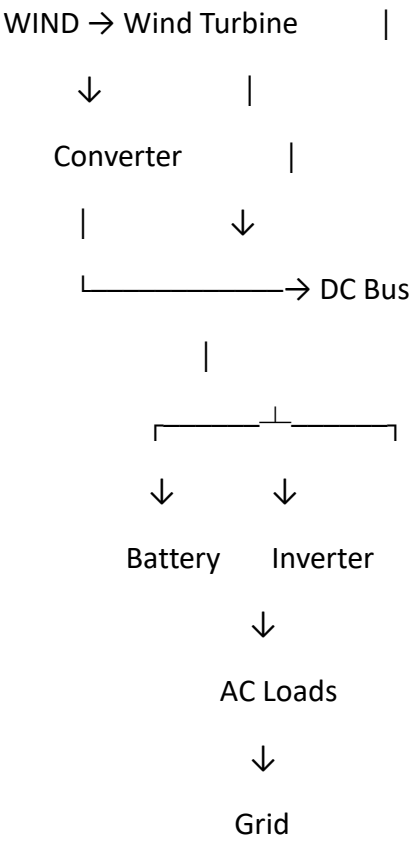
Examples:

- solar + battery
- solar + diesel
- solar + wind
- solar + hydro
- solar + biomass
- wind + hydro
- solar + wind + battery

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18.2 Solar-Wind-Battery Hybrid System





Hybrid systems can reduce the effect of resource variability.

SOLVED EXAMPLE 11: HYBRID LOAD CALCULATION

A rural health centre has:

- lighting = 1 kW for 6 h/day
- refrigerator = 0.5 kW for 10 h/day
- water pump = 2 kW for 2 h/day
- communication equipment = 0.2 kW for 8 h/day

Calculate daily energy consumption.

Lighting

$$E_1 = 1 \times 6 = 6kWh$$

Refrigerator

$$E_2 = 0.5 \times 10 = 5kWh$$

Pump

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$$E_3 = 2 \times 2 = 4kWh$$

Communication

$$E_4 = 0.2 \times 8 = 1.6kWh$$

Total:

$$E_T = 6 + 5 + 4 + 1.6$$
$$\boxed{E_T = 16.6kWh/day}$$

If system losses are 15%:

$$E_{required} = \frac{16.6}{0.85}$$
$$\boxed{E_{required} = 19.53kWh/day}$$

PART V: ENERGY PLANNING

19.1 Energy Planning

Energy planning determines:

- how much energy is required
- where it will come from
- how it will be converted
- how it will be transported
- how it will be stored
- how much it will cost
- what environmental and social effects will occur

19.2 Important Planning Questions

An engineer should ask:

1. What is the energy demand?
2. What resources are locally available?

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3. What technology is appropriate?
4. What is the capital cost?
5. What is the operating cost?
6. Is the technology reliable?
7. What environmental impacts exist?
8. What social impacts exist?
9. What infrastructure is required?
10. Can the system be maintained locally?

19.3 Energy Security

Energy security means having reliable access to sufficient, affordable and sustainable energy.

It can be improved through:

- diversified energy resources
- domestic renewable energy
- efficient infrastructure
- energy storage
- strong transmission networks
- strategic fuel reserves
- demand-side management

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PART VI: ENERGY POLICY AND THE NIGERIAN CONTEXT

20.1 Key Nigerian Energy Issues

Major issues include:

- inadequate electricity supply
- transmission constraints

- distribution losses
- dependence on fossil fuels
- rural energy access
- cooking-energy challenges
- unreliable power supply
- energy poverty
- inadequate investment
- infrastructure limitations

20.2 Opportunities for Renewable Energy in Nigeria

Solar

Excellent opportunity for:

- mini-grids
- solar home systems
- water pumping
- telecommunications
- rural electrification
- commercial rooftops

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Hydro

Suitable for:

- large hydro
- small hydro
- mini-hydro
- rural electrification

Biomass

Potential feedstocks include:

- rice husks

- cassava residues
- palm residues
- sawdust
- animal waste
- municipal waste

Wind

Potential applications include:

- water pumping
- distributed electricity
- selected grid-connected projects

PART VII: IMPORTANT ENGINEERING FORMULAE

Energy

$$E = Pt$$

Kinetic Energy

$$E_k = \frac{1}{2}mv^2$$

Potential Energy

$$E_p = mgh$$

Efficiency

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

Solar Power

$$P = GA\eta$$

Wind Power

$$P = \frac{1}{2}\rho A v^3 C_p$$

Wind Swept Area

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$$A = \pi R^2$$

Tip-Speed Ratio

$$\lambda = \frac{\omega R}{v}$$

Hydropower

$$P = \eta \rho g Q H$$

Battery Energy

$$E = V \times Ah$$

Transmission Power Loss

$$P_{loss} = I^2 R$$

Three-Phase Power

$$P = \sqrt{3} V I \cos \phi$$

Capacity Factor

$$CF = \frac{E_{actual}}{P_{rated} \times 8760}$$

Overall Conversion Efficiency

$$\eta_{total} = \eta_1 \eta_2 \eta_3 \cdots \eta_n$$

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PART VIII: ADDITIONAL SOLVED EXAMPLES

Example 12: Energy Consumption

A 150 W refrigerator operates for 12 hours per day.

Calculate monthly energy consumption for 30 days.

$$E = Pt$$
$$E = 0.15 \times 12 \times 30$$
$$\boxed{E = 54kWh/month}$$

Example 13: Solar-Battery Sizing

A household requires 10 kWh/day. The solar system receives an average of 5 peak-sun-hours per day. Assume overall system efficiency of 80%.

Required PV capacity:

$$P_{PV} = \frac{10}{5(0.8)}$$
$$\boxed{P_{PV} = 2.5kW}$$

Thus, a practical design would require approximately a **2.5 kW PV array**, subject to additional design margins and local conditions.

Example 14: Hydrokinetic Energy

For a river-current hydrokinetic turbine, the available kinetic power is:

$$P = \frac{1}{2}\rho A v^3$$

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Assume:

- water density = $1000kg/m^3$
- swept area = $5m^2$
- river velocity = $1m/s$
- $C_p = 0.35$

Then:

$$P = 0.5(1000)(5)(1)^3(0.35)$$
$$\boxed{P = 875W}$$

This illustrates why low-speed river applications require careful turbine and generator optimisation. Power varies with the **cube of velocity**.

For example, if velocity doubles:

$$P \propto v^3$$

Therefore:

$$P_2 = 2^3 P_1 = 8P_1$$

A doubling of flow velocity theoretically produces eight times the available kinetic power, before considering changes in turbine operating conditions.

PART IX: ENERGY TRANSITION

21.1 Traditional Energy System

Fossil Fuel

↓

Central Power Plant

↓

Transmission

↓

Distribution

↓

Consumer

21.2 Emerging Energy System

Solar ──┐

Wind ──┤

Hydro ──┤

Biomass ─┤

Gas ──┤

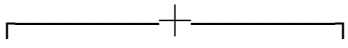
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Flexible Generation

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Transmission Grid

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Industry Transport Buildings

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↑

Storage

↑

Batteries/Hydrogen

The modern system is increasingly characterised by:

- distributed generation
- smart grids
- energy storage
- digital control
- demand response
- electric vehicles
- distributed energy resources

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PART X: SMART ENERGY SYSTEMS

22.1 Smart Grid

A smart grid uses communication, measurement, automation and control technologies to improve the operation of the electricity system.

Renewable Generation

↓

Smart Grid

↙ ↓ ↘

Homes Industry EVs

↑ ↑ ↑

└—— Data ——┘

|

Control Centre

22.2 Benefits of Smart Grids

- improved reliability
- better demand management
- renewable integration
- reduced losses
- automatic fault detection
- improved monitoring
- consumer participation

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PART XI: ENVIRONMENTAL IMPACT OF ENERGY SYSTEMS

23.1 Fossil-Fuel Impacts

Major environmental concerns include:

- CO_2
- methane
- NOx
- SOx
- particulate matter
- oil spills
- mining degradation
- water contamination

23.2 Renewable Energy Impacts

Renewable technologies also have environmental footprints.

Solar

- land use
- mining and manufacturing
- panel disposal

Wind

- visual impact
- noise
- bird/bat interactions
- material requirements

Hydro

- habitat modification
- sediment changes
- displacement
- downstream ecological impacts

Biomass

- air emissions
- land use
- deforestation risk

Therefore, environmental assessment must consider the **complete life cycle**.

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PART XII: LIFE-CYCLE THINKING

A proper energy assessment considers:

Raw Material

↓

Manufacturing

↓

Transportation

↓

Installation

↓

Operation

↓

Maintenance

↓

Decommissioning

↓

Recycling / Disposal

This is called **life-cycle assessment (LCA)**.

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PART XIII: ENERGY, CLIMATE CHANGE AND SUSTAINABILITY

Energy systems are closely linked to climate change because fossil-fuel combustion releases greenhouse gases.

A sustainable transition therefore seeks to:

1. reduce energy intensity;

2. improve efficiency;
3. electrify suitable end uses;
4. increase renewable generation;
5. develop energy storage;
6. improve grids;
7. reduce methane emissions;
8. promote sustainable fuels;
9. ensure affordability and access.

PART XIV: KEY TAKEAWAYS FOR STUDENTS

Students should understand the following central relationship:

$$Energy\ Resource \rightarrow Conversion \rightarrow Transmission \rightarrow Storage \rightarrow End\ Use \rightarrow Energy\ Service$$

The objective of engineering is not simply to produce more energy.

It is to provide **useful energy services reliably, economically, safely and sustainably**.

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REVISION QUESTIONS

Short-Answer Questions

1. Define energy.
2. Distinguish between energy and power.
3. What is primary energy?
4. What is secondary energy?
5. Define renewable energy.
6. List five renewable energy resources.
7. List five non-renewable energy resources.
8. Explain the concept of energy conversion.
9. Define energy efficiency.

10. What is energy conservation?
11. Explain the difference between renewable and non-renewable resources.
12. What is an energy mix?
13. Define capacity factor.
14. What is sustainable energy?
15. Explain energy security.
16. What is nuclear fission?
17. What is nuclear fusion?
18. Explain the photovoltaic effect.
19. State the wind-power equation.
20. State the hydropower equation.
21. What is biomass?
22. Explain anaerobic digestion.
23. What is geothermal energy?
24. Why is energy storage important?
25. What is a smart grid?

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CALCULATION QUESTIONS

Question 1

A 5 kW electric motor operates for 8 hours per day. Determine its daily energy consumption.

Question 2

A hydroelectric plant has a flow rate of 15 m^3/s, head of 40 m and efficiency of 80%. Calculate its output power.

Question 3

A wind turbine has a rotor radius of 15 m and operates at 10 m/s. If $C_p = 0.38$and air density is $1.225 kg/m^3$, calculate the mechanical output.

Question 4

A solar PV array has an area of 20 m^2, receives 700 W/m^2irradiance and has efficiency of 18%. Determine its output power.

Question 5

A 48 V, 300 Ah battery is used in an off-grid system. Calculate its nominal energy capacity.

Question 6

A 10 MW generator produces 25,000 MWh annually. Calculate its capacity factor.

Question 7

A transmission line carries 50 MW at 132 kV and power factor 0.95. Calculate the current.

Question 8

A biomass plant consumes 1,500 kg/h of biomass with a heating value of 18 MJ/kg. If electrical efficiency is 30%, calculate the output power.

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DISCUSSION QUESTIONS

1. Why should Nigeria diversify its energy mix?
2. Is renewable energy sufficient to replace fossil fuels completely?
3. Discuss the advantages and disadvantages of nuclear energy for Nigeria.
4. Why is solar energy particularly important for rural electrification?
5. Explain how energy storage can support solar and wind power.
6. Discuss the role of natural gas in Nigeria's energy transition.
7. Compare large hydropower and small hydropower.
8. Discuss the sustainability of biomass energy.
9. Explain why energy efficiency should be considered an energy resource.

10. Discuss the relationship between renewable energy and sustainable development.
11. Explain the technical, economic and social factors that influence energy-mix decisions.
12. Discuss how electric vehicles may change the future energy system.
13. Explain the importance of smart grids in renewable-energy integration.
14. Discuss the environmental impacts of hydropower.
15. Explain why renewable energy is fundamental to long-term sustainable development.

TUTORIAL: ENERGY TECHNOLOGY SELECTION

A community requires reliable electricity for:

- households
- a health centre
- schools
- water pumping
- small businesses

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Available resources are:

- strong solar radiation
- moderate wind
- a nearby river
- agricultural residues

Students should determine:

1. Which renewable technologies are suitable?
2. Should the system be standalone or grid-connected?
3. What type of storage should be used?
4. What energy mix would provide the greatest reliability?
5. What are the environmental impacts?

6. What factors determine economic feasibility?
7. How can the system contribute to sustainable development?

Recommended engineering approach

Assess Demand

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Assess Resources

↓

Select Technologies

↓

Size Components

↓

Select Storage

↓

Economic Analysis

↓

Environmental Assessment

↓

Social Assessment

↓

Optimise Energy Mix

↓

System Implementation

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SUMMARY OF THE ENTIRE COURSE

The major lesson of GEE 306 is that the future energy system will not be based on one technology.

Instead, a sustainable energy system will involve an **integrated energy mix** combining:

- renewable energy
- appropriate conventional generation
- energy storage
- efficient transmission
- smart grids
- energy efficiency
- sustainable end-use technologies
- responsible energy policies

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Globally, renewable energy is expanding rapidly. IRENA's latest statistics show that global renewable capacity reached about **5.15 TW at the end of 2025**, after approximately **692 GW** of additions during the year. (IRENA)

For Nigeria, the engineering challenge is somewhat broader than simply replacing fossil fuels. The country must simultaneously improve **energy access, reliability, affordability, economic development and environmental sustainability**. This is reflected in Nigeria's Energy Transition Plan and its 2060 net-zero pathway. (Nigeria Energy Transition Plan)

The fundamental principle can therefore be stated as:

Sustainable Energy = Reliable + Affordable + Accessible + Efficient + Environmentally Responsible

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